A simpler proof of Lemma 1, discovered by an AI-assistant within the Reaslab[1] toolkit.

Lemma 1. *Suppose that G is a graph, E is a subset of graph edges, $p_1, \ldots, p_m$, where m is odd, are acyclic paths connecting vertices s and t. Additionally suppose that any edge from E is contained in an even number of paths p_i. Then the vertices s and t are connected by a path composed of segments of the paths p_i, and this path does not contain any edge from E.*

Proof. Let us introduce an indicator function $\chi(v)$ for the reachability of a vertex $v \in G$ from s via a path that avoids E. In particular, $\chi(s) = 1$. Assume the contrary—that vertex t is not reachable from s by such a path. This implies that $\chi(t) = 0$.

For an edge $e = (u, v)$, let us define $\chi(e) = \chi(u) + \chi(v) \bmod 2$. Clearly, for any edge $e = (u, v) \notin E$, we have $\chi(u) = \chi(v)$, and therefore $\chi(e) = 0$.

On the one hand, given that the number of paths is odd, we have

$$\chi := \sum_{i=1}^{m} \sum_{e \in p_i} \chi(e) \equiv m(\chi(s) + \chi(t)) \equiv m \equiv 1 \pmod 2.$$

For an edge $e \in E$, let $\sigma(e)$ denote the number of paths p_i that pass through e. Then, given that all $\sigma(e)$ are even and the paths are acyclic (each path traverses any given edge at most once), we have

$$\chi \equiv \sum_{e \in E} \sigma(e)\chi(e) \equiv 0 \pmod 2.$$

A contradiction. □

[1] https://reaslab.io